

Typed Budgets for Object-Relational Driving World Models

Peinan Li, Zhiwen Chen, Yibo Song, Ruiyang Gao, Yixiang Yang, Ke Wang, and Zhipeng Zhang*

Abstract—While object-relational world models provide crucial interaction cues for autonomous driving policies, their fixed-capacity latent representations necessitate a stringent token selection mechanism. Existing methods typically force relation and dynamic agent tokens to compete within a single top- K pool. This creates a critical failure mode where relation tokens frequently displace essential agents that the driving policy must track. To address this limitation, we formulate token selection as a budget allocation problem and propose DOOR. The architecture allocates separate token budgets for dynamic agents and relational interactions before integrating them into a shared world model. Under a strict 16-slot budget on the nuScenes dataset, our approach reduces rare-agent displacement error by 55.2% and improves Interaction Recall@1m by 8.3 points compared to object-only baselines, whereas naive token mixing suffers severe degradation. We also evaluate downstream policy performance. On the planning-oriented nuPlan benchmark, our no-visibility decoupled variant outperforms the object-only baseline at the 50k scale, achieving a higher return (14.51 ± 2.93 vs. 1.72 ± 17.89) and a lower imagined collision rate (0.259 ± 0.045 vs. 0.610 ± 0.146). Offline planner-like evaluations further support these gains with reduced teacher-action MSE (6.63 ± 0.11 vs. 8.86 ± 0.37). Cross-dataset rollouts and horizon sensitivity analyses show that typed budget allocation improves latent representations, but policy improvements are most pronounced when the downstream evaluation effectively leverages relational cues.

Index Terms—Autonomous driving, world models, object-relational representation, policy learning.



1 INTRODUCTION

WORLD models have emerged as a transformative paradigm for autonomous driving, enabling policies to train on imagined future trajectories rather than relying exclusively on costly, risk-sensitive, and sample-inefficient environmental interactions [1], [2]. However, driving is fundamentally a multi-agent interactive game governed by strict physical and topological constraints. To make safe and informed decisions, a driving policy cannot rely on an unstructured, flat latent embedding that entangles all scene elements. Instead, it must explicitly track a critical subset of dynamic agents (e.g., vehicles, pedestrians) alongside the relational structures (e.g., time-to-collision, lane conflicts, right-of-way) that dictate their interactions.

This compositional requirement introduces a fundamental architectural challenge: *how should relation-aware structures be integrated into a fixed-capacity latent driving model?* Due to computational and real-time constraints, world models must compress the structured scene into a compact latent bottleneck (e.g., a top- K slot selection). A prevalent, albeit naive, approach is to append relation tokens to the same candidate pool as dynamic agents, operating under the assumption that an attention mechanism or a shared scoring network can optimally select the most critical tokens regardless of their semantic type.

In the context of autonomous driving, we identify this shared-bottleneck assumption as highly brittle. Relational

features are inherently dependent on the entities they describe. A time-to-collision warning or a lane-conflict indicator provides actionable value to the policy only if the specific agents involved in that relation are simultaneously preserved within the latent state. When forced to compete in a single top- K pool, high-risk relation tokens frequently crowd out the very dynamic agents they refer to. Consequently, the latent state may contain warnings about an impending interaction while missing the physical agent states needed to imagine the effect of an ego action.

We formalize this issue as an *inter-type budget competition* problem. The efficacy of relation-aware driving world models depends entirely on how representational slots are allocated across heterogeneous token families. To address this, we propose DOOR, a typed-budget object-relational abstraction. DOOR strictly decouples the selection process, allocating separate token budgets for dynamic agents and relational interactions before feeding the aggregated slots to a shared reactive world model. Our core hypothesis is that typed budgeting preserves decision-critical dynamic coverage while retaining a reserved capacity for the highest-risk relational cues, all without expanding the total latent context size.

Beyond architectural design, this paper investigates a deeper scientific question: *does an enhanced relational representation inherently translate to improved policy learning?* Better predictive metrics in a world model do not automatically simplify latent policy optimization. To rigorously decouple these effects, we design a two-stage evaluation protocol. Stage 0 measures representation sufficiency, specifically dynamic forecasting and rare-agent tracking, under a strict 16-slot budget on the nuScenes dataset. Stage 1 then evaluates imagination-based policy learning across both the short-

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horizon nuScenes [3] benchmark and the denser, planning-oriented nuPlan [4] benchmark at the 20k and 50k scales.

Our comprehensive evaluation reveals a fundamental decoupling between representation quality and downstream policy utility. In Stage 0, DOOR avoids the catastrophic failure mode of naive mixing, reducing rare-agent displacement error by 55.2% and improving Interaction Recall@1m by 8.3 points. However, Stage 1 demonstrates that policy gains are highly context-dependent. On nuScenes, where the evaluation horizon is short and relational cues are less critical for the specific metrics, object-only representations remain the strongest baseline. Conversely, on the complex nuPlan benchmark, the ranking flips. Our decoupled no-visibility variant outperforms object-only baselines, achieving a higher return (14.51 ± 2.93 vs. 1.72 ± 17.89) and a lower imagined collision rate (0.259 ± 0.045 vs. 0.610 ± 0.146) at the 50k scale. Offline planner-like evaluations and interaction-conditioned subset analyses further confirm that DOOR excels precisely in scenarios dominated by lane conflicts and low time-to-collision events.

The scope of this study is deliberately focused. Rather than presenting a monolithic perception-to-control system, we study decision-sufficient abstraction using oracle tokens. This targeted approach isolates the representation-policy interface, removing perception noise to definitively answer how latent capacity allocation affects downstream control.

In summary, the main contributions of this work are:

- 1) We identify and formalize the phenomenon of *inter-type budget competition* in relation-aware driving models, demonstrating that naively mixing relation tokens into a shared bottleneck severely degrades dynamic-agent coverage and can harm downstream policy learning.
- 2) We propose DOOR, a typed-budget object-relational abstraction that explicitly decouples dynamic-agent and relation-token selection, ensuring decision-sufficient context preservation under a strict latent capacity limit.
- 3) Through a rigorous two-stage evaluation, we reveal a critical insight into world-model-based driving: typed budgeting improves latent representation sufficiency, while policy gains materialize primarily when the downstream environment possesses sufficient interactive complexity to exploit relational tokens.

2 RELATED WORK

2.1 World Models for Driving Policy Learning

World models train policies through learned latent roll-outs rather than direct environment interaction [1], [2], [5], [6], [7]. The idea is rooted in model-based reinforcement learning and planning-through-simulation, from Dyna-style updates to modern latent planners and model-based policy optimization [8], [9], [10], [11], [12]. Driving makes this appealing because online trial-and-error is costly and risk-sensitive. Model-based driving systems and imitation frameworks use learned dynamics, compact scene states, or planner-like supervision to reduce online interaction [13], [14], [15], [16]. These works usually keep the latent scene

interface fixed. Here the interface itself is under study: we ask whether the world model is given a decision-sufficient set of entities and relations before any policy update occurs.

2.2 Object-Centric and Relational Representations

Object-centric learning represents scenes using discrete slots or entities [17], [18]. Furthermore, relational inductive biases and graph reasoning explicitly model interactions, rather than relying on implicit inference from pooled embeddings [19], [20], [21]. Autonomous driving policies inherently require similar structural priors. Ego actions are continuously shaped by physical entities such as nearby vehicles and vulnerable road users, alongside relational factors including map constraints, right-of-way, lane conflicts, and time-to-collision. Earlier multi-agent prediction models already showed the value of interaction-aware futures and social context [22], [23], [24], [25]. To expose these interactions and map structures, forecasting systems such as VectorNet, LaneGCN, TNT, DenseTNT, HiVT, and MultiPath++ employ vectorized agents, lane graphs, discrete goals, and multimodal futures [26], [27], [28], [29], [30], [31]. Recent attention-based predictors further scale this paradigm using agent-map transformers, query-centric decoding, and compact attention patterns [32], [33], [34], [35], [36]. In contrast to these architectural advancements, our work addresses a more specific bottleneck. We investigate how pre-extracted object and relation tokens should be integrated into a fixed-capacity world model.

2.3 Planning-Oriented Autonomous Driving

Planning-oriented driving systems organize perception, prediction, and planning around structured scenes [37], [38], [39]. These architectures build upon foundational methods that utilize rasterized, vectorized, occupancy-centric, or fused sensor encodings to generate ego trajectories subject to map and actor constraints [40], [41], [42], [43], [44]. Object-level planners explicitly address this design pressure by providing the planner with direct access to agent states and route contexts [45]. While full end-to-end evaluations are comprehensive, they inherently conflate perception errors, supervision signals, simulator dynamics, and planning capabilities. To circumvent these confounding factors, our study utilizes oracle tokens to isolate a specific component of the autonomy stack. This targeted approach enables us to rigorously evaluate whether a typed allocation of dynamic-agent and relation-token capacity enhances latent world-model learning and successfully transfers to downstream policy optimization.

2.4 Driving Benchmarks and Cross-Dataset Evaluation

nuScenes [3], Argoverse [46], [47], Waymo Open Motion [48], [49], and nuPlan [4] stress different parts of the driving stack. CARLA [50], Waymax [51], and NAVSIM [52] make a related point from the simulator side: the downstream task changes the meaning of a representation. A representation that improves prediction on one dataset may not improve policy learning on another, especially when token statistics, visibility fields, teacher-action scales, and interaction density differ. We use this contrast by evaluating the same abstraction family on nuScenes short-horizon imagination and nuPlan planning-oriented imagination.

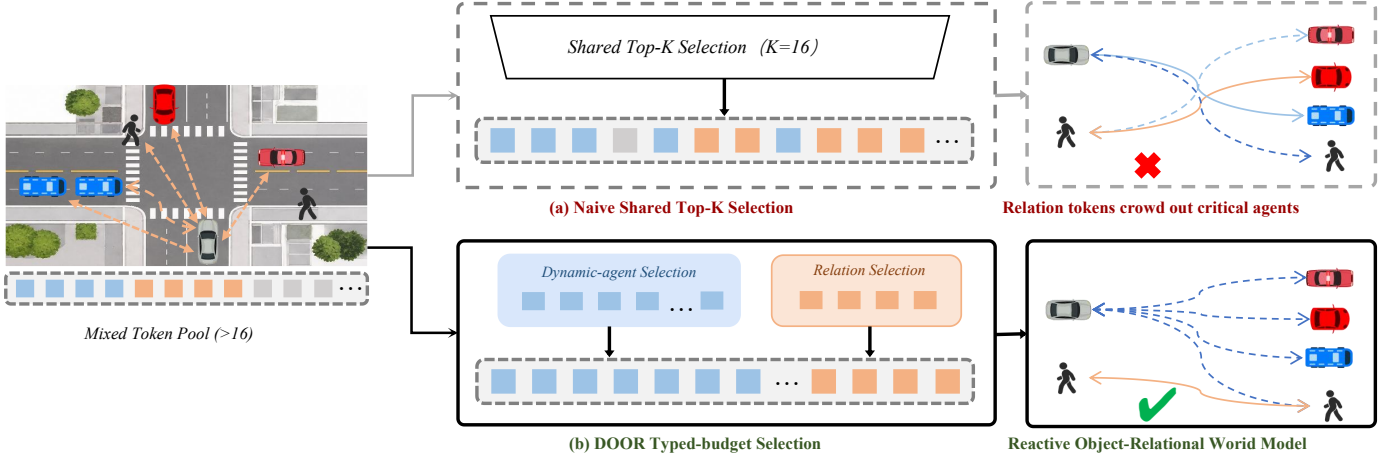


Fig. 1. Overview. Under a fixed 16-slot latent budget, naive shared object-relation selection can displace decision-critical dynamic agents. Typed-budget decoupling restores that context, and the policy benefit appears only when the evaluation can use relation structure.

3 PROBLEM FORMULATION

We consider a driving agent acting in a partially observable traffic environment. At time step t , the agent receives a structured scene representation

$$\mathbf{x}_t = \{\mathbf{e}_t, \mathcal{O}_t, \mathcal{M}_t, \mathcal{R}_t\}, \quad (1)$$

where $\mathbf{e}_t$ denotes the ego token, $\mathcal{O}_t$ denotes dynamic-agent tokens, $\mathcal{M}_t$ denotes map tokens, and $\mathcal{R}_t$ denotes relation tokens. The policy $\pi(\mathbf{a}_t | \mathbf{x}_t)$ outputs a driving action $\mathbf{a}_t$, which may correspond to low-level control or a compact motion command.

Since the world model cannot consume the full scene, it compresses the input $\mathbf{x}_t$ into a latent context of fixed capacity K . Consequently, the fundamental challenge lies not only in *what* to encode, but also in *how* to allocate this limited capacity between dynamic-agent and relational information. If relation tokens compete within the same bottleneck as dynamic-agent tokens, they risk displacing the very entities crucial for control. To address this, the abstraction A_θ and the world model f_θ are trained to ensure that the latent state remains decision-sufficient for two distinct objectives: (1) **representation sufficiency**, which preserves dynamic prediction, rare-agent tracking, and interaction-critical structures under a fair fixed latent budget; and (2) **policy learning**, which enables stable, imagination-based policy optimization.

3.1 Scope and Evaluation Assumptions

The experiments study decision learning with structured scene tokens rather than full perception. In *oracle-token evaluation*, scene tokens come from annotations, simulator state, or an equivalent trusted front-end. This removes perception noise from the abstraction-and-policy question.

We use two criteria for a decision-sufficient abstraction. First, the selected context should preserve entities that the policy may need to react to, especially near-field dynamic agents and rare vulnerable road users. Second, it should preserve compact interaction evidence, such as low time-to-collision or lane conflict, without letting those relation tokens remove the entities that make the relation meaningful. A representation can satisfy one criterion and fail the other. Object-only selection keeps dynamic agents but lacks

TABLE 1
Raw token fields used by the structured scene interface.

Dim.	Field	Notes
0–1	(x, y)	ego-centric relative position
2–3	(v_x, v_y)	relative or local-frame velocity
4–5	length, width	object extent when available
6	risk	inverse-distance or adapter risk proxy
7	visibility	nuScenes bin midpoint; nuPlan is 1.0
8	TTC	relation-token time-to-collision proxy
9	lane conflict	1 if lateral offset is below 2 m
10	priority	interaction-priority heuristic
11	distance	ego-object distance
12	heading	local heading
13	interactive flag	reserved interaction marker
14	token type	ego/object/map/relation/pad id

explicit interaction semantics; naive object-relation selection keeps relation tokens but can lose dynamic coverage. Typed budgets are designed to make this trade-off explicit.

4 METHOD

4.1 Scene Tokenization

Each scene is represented in an ego-centric local frame to encourage cross-benchmark consistency. The raw feature layout used by the current implementation contains a fixed-length token vector of dimension $D_{\text{raw}} = 40$, with the first coordinates reserved for relative position, velocity, object extent, risk proxies, visibility, time-to-collision, lane conflict, interaction priority, distance, heading, and token type. The maximum token budget is $S = 97$, corresponding to one ego token, a bounded set of dynamic-agent tokens, map tokens, relation tokens, and padding.

The design keeps the interface stable across datasets and relation features explicit. Relation tokens encode relative geometry, relative velocity, time-to-collision proxy, lane conflict, visibility, and interaction priority. Both Stage-1 settings share this schema, yet token statistics differ since nuScenes contains visibility variation while nuPlan yields nearly constant visibility. This impacts visibility ablations.

Table 1 summarizes the feature fields used by the implementation. Only the first 15 dimensions are semantically assigned; the remaining dimensions are reserved so that additional map, signal, or dataset-specific attributes can be

added without changing the model interface. The ego token always occupies slot 0. Dynamic-agent tokens are appended next, followed by map-anchor tokens, relation tokens, and padding. The nuPlan preprocessed adapter uses route-lane anchors before other lane anchors for map tokens, whereas the nuScenes adapter uses available local map information. Relation tokens are ego-object edges: for every dynamic object we compute relative displacement, relative velocity, time-to-collision, inverse-distance risk, a lane-conflict indicator, visibility, and priority. If more relation edges are available than the relation-token cap, they are sorted by risk and the highest-risk edges are retained.

For nuScenes, visibility tokens are converted from the official four-bin annotation into continuous midpoints. For nuPlan, the preprocessed pipeline does not provide comparable visibility variation, so the field is constant at one. Thus the no-visibility ablation removes different information on the two datasets.

4.2 Decision-Sufficient Abstraction

Given token features $\mathbf{x}_{t,i}$ and token types $c_{t,i}$, an encoder produces latent token embeddings

$$\mathbf{h}_{t,i} = g_{\text{enc}}(\mathbf{x}_{t,i}, c_{t,i}), \quad (2)$$

where g_{enc} is implemented as a token encoder with learned type embeddings. A decision-sufficient abstraction module then scores tokens using ego-conditioned attention:

$$\alpha_{t,i} = \text{softmax}_i \left(\frac{(\mathbf{W}_q \mathbf{h}_{t,\text{ego}})^\top (\mathbf{W}_k \mathbf{h}_{t,i})}{\sqrt{d}} \right), \quad (3)$$

followed by a learned importance head

$$s_{t,i} = g_{\text{score}}([\mathbf{h}_{t,i}; \alpha_{t,i}]). \quad (4)$$

The top- K tokens form the selected latent entity set $\mathbf{z}_t^{\text{sel}}$. In the current implementation, $K = 16$. A pooled global latent $\mathbf{z}_t^{\text{glob}}$ is computed in parallel for policy and value prediction.

This abstraction is the decision bottleneck. If it drops interaction-critical structure, the policy should degrade in the scenes where those interactions matter.

For the Stage-0 representation study, we decouple this bottleneck by token type. Instead of forcing dynamic-agent and relation tokens to compete inside one top- K pool, we allocate typed budgets K_{dyn} and K_{rel} such that

$$K_{\text{dyn}} + K_{\text{rel}} = K. \quad (5)$$

The world model receives the concatenated dynamic and relation slots. Under the fair 16-slot budget used in our experiments, we set $(K_{\text{dyn}}, K_{\text{rel}}) = (12, 4)$. This targets a failure mode of shared selection: relation tokens can displace dynamic-agent slots needed for forecasting and interactive decisions. Typed budgeting keeps the total context size fixed while avoiding inter-type competition.

The decoupled path uses two instances of the same scoring module. The dynamic path selects from ego, vehicle, pedestrian, and cyclist tokens, with the ego token forced into the selected set. The relation path selects only relation tokens and does not force-select ego, which would waste one relation slot. Dynamic slots are concatenated before relation slots, so the downstream loss and evaluator can recover slot type through the original token indices. In the default decoupled condition, the policy input averages the

dynamic and relation global latents. In the relation-to-critic-only check, the actor receives only the dynamic latent, while the critic receives the combined latent. Scenes with fewer than K_{rel} valid relations produce masked relation slots, ignored by the transformer mask and by all losses.

The (12, 4) split is not universal. It is the reference split for the fair 16-slot study, providing enough dynamic capacity for the local agent set and a small relation budget for highest-risk edges. A relation-heavy (10, 6) split in Table 4 still works but is weaker, indicating that allocation matters beyond the presence of relation tokens.

The typed split acts as a weak structural prior. It does not prescribe which vehicle, pedestrian, cyclist, or relation to select. These choices remain learned and data-dependent. It only prevents one semantic family from exhausting the compact context. This matters because relation tokens are often numerous and high-scoring under risk heuristics. In a shared top- K pool, a cluster of relation tokens for one interaction can crowd out several nearby agents. The resulting latent state may warn of an interaction while missing the agent state needed to imagine ego actions. Typed selection reduces this mismatch by requiring a minimum dynamic-agent inventory before adding relation evidence. The design is deliberately conservative: it changes the allocation rule, not the encoder, world model, or policy optimizer. Figure 2 contrasts shared top- K selection with typed-budget decoupling inside the same world-model pipeline.

4.3 Reactive Object-Relational World Model

The world model receives the selected token set $\mathbf{z}_t^{\text{sel}}$ and action $\mathbf{a}_t$, then predicts next-step latent entities, reward, continuation probability, and collision risk:

$$\hat{\mathbf{z}}_{t+1}^{\text{sel}}, \hat{r}_t, \hat{\rho}_t, \hat{c}_t = f_\theta(\mathbf{z}_t^{\text{sel}}, \mathbf{a}_t). \quad (6)$$

The action enters the token dynamics, so predicted transitions depend on ego intervention. Without this conditioning, the latent model would behave more like a replay predictor than a reactive model.

The action is projected to the model dimension and prepended as an action token. A two-layer transformer encoder processes the action-token-plus-slot sequence with key padding for masked slots. Transformed non-action slots are decoded to the raw 40-dimensional token space; a masked mean feeds reward, continuation, and collision heads. All fair variants use this world model and differ only in which 16 slots reach it.

The training loss combines next-token prediction, reward prediction, continuation prediction, and auxiliary collision-risk prediction:

$$\mathcal{L}_{\text{wm}} = \lambda_{\text{obs}} \mathcal{L}_{\text{obs}} + \lambda_r \mathcal{L}_r + \lambda_\rho \mathcal{L}_\rho + \lambda_{\text{col}} \mathcal{L}_{\text{col}}. \quad (7)$$

The current scaffold also allows an optional behavior-cloning regularizer during warm start,

$$\mathcal{L} = \mathcal{L}_{\text{wm}} + \lambda_{\text{bc}} \mathcal{L}_{\text{bc}} + \mathcal{L}_{\text{pol}}, \quad (8)$$

which stabilizes early training when the world model and value function are still inaccurate.

The observation loss is type-aware. Dynamic slots, including ego, vehicles, pedestrians, and cyclists, are supervised only on (x, y, v_x, v_y) . Relation slots are supervised only on TTC, lane-conflict, and priority fields. Map, signal,

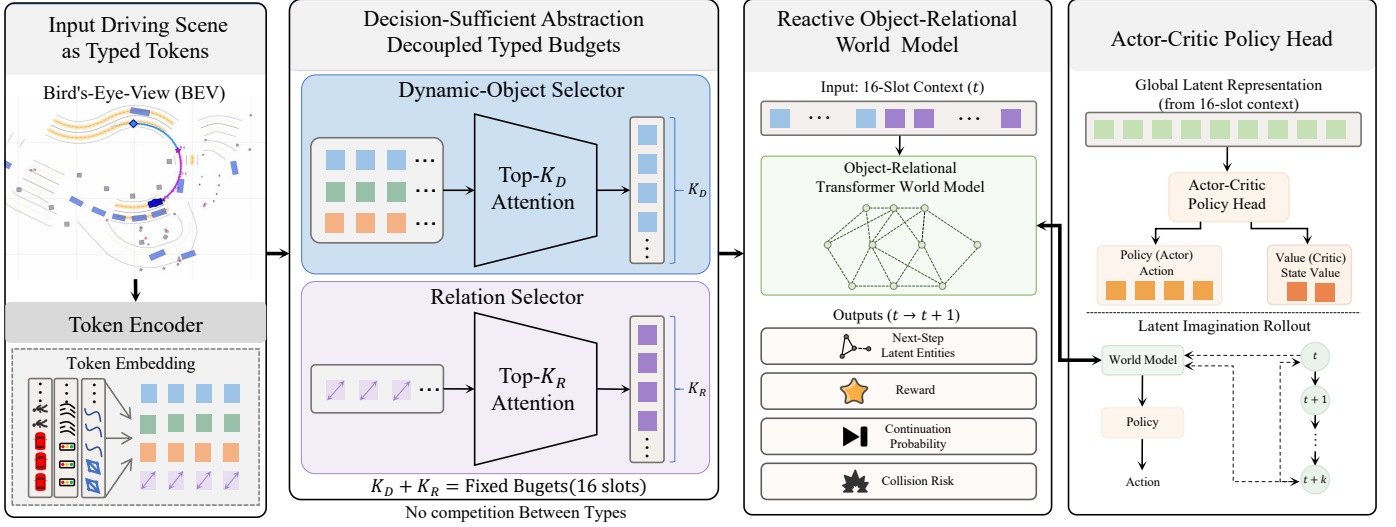


Fig. 2. DOOR pipeline. Naive shared-budget selection is compared with typed dyn/rel selection; the fair (12, 4) split sits inside the same 16-slot budget, and selected slots feed the action-conditioned world model and policy/value update.

and padding slots do not contribute to the next-token observation loss. This avoids forcing relation slots to regress non-physical position or extent targets and aligns the training objective with the representation metrics. For holistic learned-query slots, which no longer correspond to original token indices, we use a set-prediction loss. Each ground-truth dynamic agent is matched to the nearest predicted slot by (x, y) , and the loss is applied on the matched (x, y, v_x, v_y) fields. Collision targets are the same for training and evaluation: a sample is positive if any valid relation token has TTC below 3s.

The model still has to learn cross-type dependencies. Relation slots are not decoded in isolation: they pass through the same transformer as dynamic slots, so the predicted next dynamic state can attend to selected relation evidence and the predicted relation fields can attend to selected agent states. The typed budget changes which tokens enter this transformer, not the transformer itself. This distinction is useful for downstream interpretation. If typed budgeting improves Stage-0 dynamic rollout, it suggests that the selected context is more usable by the same dynamics model; if it fails in Stage 1, the bottleneck improvement did not translate into easier policy optimization.

4.4 Latent Imagination Policy Learning

Policy and value heads are trained on imagined trajectories from the world model. At each rollout step, the policy head reads $\mathbf{z}_t^{\text{glob}}$ and outputs a diagonal-Gaussian action distribution. The action mean is tanh-bounded and the log standard deviation is clamped before sampling. A value head reads the same global latent. Stage 1 uses a 5-step rollout, GAE- λ targets, reward clipping, detached world-model updates, and a real-batch consistency loss. This tests whether the Stage-0 representation gain helps policy learning.

The action schema is two-dimensional: longitudinal velocity along the ego heading and yaw rate. For nuPlan, these are derived from the first future ego pose in the pre-processed NPZ sample; for nuScenes, the adapter uses the available ego-motion or CAN-bus-derived teacher action. During imagination, selected-slot predictions are scattered

Algorithm 1 DOOR Policy-Learning Procedure

- 1: Collect or load structured scene transitions from offline data and reactive rollouts.
- 2: Encode tokens and compute decision-sufficient latent entities.
- 3: Update the world model using $\mathcal{L}_{\text{wm}}$ in (7).
- 4: Warm start the policy with behavior cloning if expert actions are available.
- 5: Roll out imagined latent trajectories with the current policy and world model.
- 6: Update the policy and value heads using imagined rewards, continuation, and value targets.
- 7: Evaluate the policy on the primary held-out validation protocol and supplementary checks.
- 8: Keep the final model according to the paper’s primary validation protocol.

back into the original 97-slot token layout, non-selected tokens are kept fixed, and the scene is re-encoded before the next abstraction step. This token-space rollout is slower than a pure latent rollout, but it ties Stage 1 to the same encoder and selector used in Stage 0. For learned-query holistic slots, original token identities are unavailable, so the rollout uses a frozen-token approximation; holistic results are therefore treated as secondary checks rather than central evidence.

Stage 1 uses the following shaped reward:

$$r_t = w_{\text{prog}} v_{x,t+1}^{\text{ego}} - w_{\text{coll}} \sigma(\hat{c}_t) - w_{\text{act}} \|\mathbf{a}_t\|_2^2, \quad (9)$$

with $(w_{\text{prog}}, w_{\text{coll}}, w_{\text{act}}) = (1.0, 5.0, 0.01)$ and per-step clipping to $[-5, 5]$. The world-model reward head is added to this shaped reward, but in offline driving data it contributes weakly because no dense RL reward is available. Policy optimization uses GAE with $(\gamma, \lambda) = (0.97, 0.95)$, an entropy term, a Huber critic loss, and a Stage-0 real-batch consistency loss. Stage 1 is a controlled representation-to-policy test, not a new driving reward.

Algorithm 1 summarizes the training procedure. Policy improvement stays in latent space, letting us test whether the abstraction changes downstream optimization.

TABLE 2
Experimental protocols.

Protocol	Dataset and Split	Purpose
Stage 0	nuScenes, 700 scenes, 28 096 samples	Representation sufficiency
Stage 1A	nuScenes, 700 scenes, 5 622 val samples	Short-horizon imagination training
Stage 1B	nuPlan 20k, 16k/4k split	Planning-oriented imagination training
Stage 1C	nuPlan 50k, 40k/10k split	Scale-up and planner-like evaluation

5 EXPERIMENTAL SETUP

5.1 Datasets and Protocols

Table 2 lists the protocols. Stage 0 uses nuScenes for representation analysis under a fair 16-slot world-model budget. Stage 1 reuses the same abstraction family for imagination-based policy learning on nuScenes and on nuPlan pre-processed benchmarks at 20k and 50k scale. The strongest nuPlan condition is then evaluated with an offline planner-like check on the 50k validation split and with interaction-conditioned subsets over lane-conflict, low-TTC, and dense-agent scenes. Cross-dataset rollouts and a nuScenes horizon sweep provide additional checks. The nuPlan Stage-1 experiments use Stage-0 warm starts trained on the corresponding dataset split.

The nuScenes protocol uses 700 scenes from the trainval split after CAN-bus and invalid-value filtering, yielding 28 096 tokenized samples. The split is scene-level 80/20, with 560 training scenes and 140 validation scenes, to avoid temporal leakage between train and validation frames from the same scene. For nuPlan, we use preprocessed NPZ samples with a 2-D teacher action derived from the next ego pose: longitudinal velocity along the ego heading and yaw rate over the first 0.1 s future step. The 20k protocol uses a 16k/4k split, and the 50k protocol uses a 40k/10k split. The 50k index is a balanced subset used consistently across the Stage-1 scale-up, planner-like evaluation, relation feature ablation, and interaction-conditioned subset analysis.

All comparisons keep the tokenization, raw feature dimension, world-model architecture, optimizer family, and compact slot budget fixed unless the row explicitly states otherwise. The intent is to compare allocation policies rather than model capacity. The 97-token reference is therefore not treated as a fair competitor; it is an upper-bound check showing what happens when the bottleneck is mostly removed. Likewise, the holistic learned-query model is informative but not central, because it changes token identity and makes relation-aware rollout less direct.

We report averages over seeds whenever repeated runs are available and use the same validation split for all variants inside a protocol. For Stage 0, the primary comparison is the fair 16-slot setting. For Stage 1, the primary comparison is within each dataset-scale block rather than across datasets, because the reward scale and teacher-action distribution differ between nuScenes and nuPlan. Cross-dataset transfer rows are diagnostic: they test whether a warm-started abstraction can survive a dataset shift, but they are not used to claim a universal ranking.

5.2 Variants

Stage 0 compares six compact 16-slot variants together with one 97-token upper-bound reference:

- 1) **Holistic-16Slot**: learned-query compression from the full token set.

- 2) **Object-Only-16**: top- K selection over ego and dynamic-agent tokens only.
- 3) **Object+Relation-16**: naive shared-budget top- K over dynamic and relation tokens.
- 4) **Object+Relation+Visibility-16**: shared-budget relation-aware selection with visibility weighting.
- 5) **Obj+Rel-Decoupled**: typed-budget abstraction with 12 dynamic slots and 4 relation slots.
- 6) **Decoupled+Visibility**: typed-budget abstraction plus visibility weighting on the dynamic path.

Stage 1 centers on `wm_object`, `wm_decoupled`, and `wm_decoupled_no_vis`. We also report a shared-relation 20k baseline (`wm_naive`), a 50k visibility arm (`wm_decoupled(+vis)`), a low-cost bc planner-imitation anchor, and two nuScenes-only checks: 14+2 typed budget and relation-to-critic-only fusion.

The Stage-1 names map directly to the Stage-0 abstractions. `wm_object` warm-starts from Object-Only-16. `wm_decoupled` uses the decoupled typed-budget model with visibility weighting on the dynamic path. `wm_decoupled_no_vis` uses the same typed-budget split but disables the visibility multiplication. The 20k `wm_naive` row warm-starts from the shared Object+Relation abstraction, so it tests whether the shared-relation mechanism remains weak once policy learning is added.

5.3 Metrics

Stage 0 reports four representation-centric metrics.

- 1) **Dyn Rollout MSE**: nearest-match squared error on (x, y, v_x, v_y) for dynamic agents.
- 2) **Collision F1**: binary F1 of predicted collision risk against labels derived from relation-token time-to-collision below 3 s.
- 3) **Rare ADE**: nearest-match displacement error on pedestrians and cyclists.
- 4) **Interaction Recall @ 1 m**: fraction of rare agents within 20 m of ego whose nearest-match error is below 1 m.

Stage 1 reports imagination-policy metrics:

- 1) **Return**: K -step imagined latent reward sum.
- 2) **Collision Rate**: fraction of imagined rollouts whose collision probability exceeds 0.5.
- 3) **Collision Mean**: mean of the maximum predicted collision probability over the imagined rollout.
- 4) **Ego Stability (cos)**: cosine distance between successive ego latents, used to flag liveness and divergence rather than to rank models.

For the nuPlan 50k planner-like evaluation, we also report teacher-action mean-squared error and action ΔL_2 distance. The same Stage-1 metrics are reused for cross-dataset transfer and horizon sensitivity.

For Stage 1, return is the undiscounted sum of the five imagined per-step rewards in (9). Collision rate is the fraction of validation rollouts whose maximum predicted collision probability over the horizon exceeds 0.5; collision mean is the mean of that same maximum probability. Ego stability is the mean step-to-step cosine distance of the ego-slot latent over the imagined rollout. The quantity is used

TABLE 3

Stage-0 nuScenes representation ablation under a fair 16-slot context budget. Results are averaged over three seeds; $\pm$ is sample standard deviation. Bold marks the best fair-budget result in each metric column. Δ rows are computed from unrounded run means.

Representation	Ctx	Dyn Rollout $\downarrow$	Collision F1 $\uparrow$	Rare ADE $\downarrow$	IntRec@1m $\uparrow$
Holistic-16Slot	16	2.11 ± 0.16	0.978 ± 0.011	1.42 ± 0.01	0.643 ± 0.015
Object-Only-16	16	3.74 ± 1.01	0.946 ± 0.004	1.10 ± 0.12	0.901 ± 0.034
Object+Relation-16 (naive)	16	40.28 ± 29.54	0.980 ± 0.013	7.51 ± 5.48	0.430 ± 0.407
Object+Relation+Visibility-16	16	15.80 ± 9.93	0.933 ± 0.064	2.96 ± 1.64	0.728 ± 0.155
Obj+Rel-Decoupled (Ours)	16	2.11 ± 0.19	0.929 ± 0.039	0.49 ± 0.18	0.984 ± 0.014
Decoupled+Vis (Ours)	16	1.88 ± 0.23	0.926 ± 0.029	0.52 ± 0.05	0.979 ± 0.008
Δ Obj+Rel-Decoupled vs Object-Only	–	–43.5%	–1.8 pts	–55.2%	+8.3 pts
Δ Decoupled+Vis vs Object-Only	–	–49.9%	–2.1 pts	–52.6%	+7.8 pts

as a warning signal rather than a ranking metric: very high values indicate possible latent drift, while very low values can also indicate a collapsed or inactive policy.

5.4 Implementation Details

The shared model configuration uses raw token dimension 40, model dimension 128, hidden dimension 256, four attention heads, two transformer world-model layers, dropout 0.1, and a token budget of 97. The fair compact setting uses $K = 16$ latent slots. Stage 0 uses Adam with learning rate 10^{-3} , weight decay 10^{-5} , batch size 32, 15 epochs, and three seeds $\{7, 42, 2026\}$. The Stage-0 loss weights are $\lambda_{\text{obs}} = 1.0$, $\lambda_r = 0.5$, $\lambda_p = 0.25$, $\lambda_{\text{col}} = 0.25$, and $\lambda_{\text{bc}} = 0.1$.

Stage 1 uses a 5-step imagination horizon, batch size 128, learning rate 4×10^{-3} , 10 epochs, three seeds $\{7, 42, 123\}$, GAE- λ targets with $(\gamma, \lambda) = (0.97, 0.95)$, reward clipping to $[-5, 5]$, a Huber critic with delta 10, a real-batch consistency loss weight of 1.0, and detached world-model dynamics during actor-critic updates. The third Stage-1 seed differs from Stage 0 because these policy runs were launched as a separate actor-critic sweep. The diagonal-Gaussian actor uses a tanh-bounded mean in $[-3, 3]$, clamps log standard deviation to $[-2, 0.5]$, and clips actions during training.

The offline planner-like evaluation is deliberately simple. It does not run a closed-loop simulator or invoke an external planner. Instead, it compares the learned policy action against teacher-derived ego actions on held-out nuPlan samples and reports the same collision-oriented imagination metrics. This makes the check reproducible and tied to the paper’s representation question, but it should not be read as a substitute for official nuPlan closed-loop scoring.

6 RESULTS

6.1 Stage-0 Representation Sufficiency on nuScenes

Table 3 gives the main representation ablation. The question is how relation structure should enter a fixed-capacity world model: relation tokens may improve forecasting, or they may consume slots needed for dynamic agents. The table reports the fair 16-slot comparison. The 97-token holistic model is kept as a non-fair upper bound.

Under the same 16-slot budget, typed-budget variants are best on interaction-critical representation metrics. Relative to Object-Only-16, Obj+Rel-Decoupled reduces Rare ADE by 55.2% and improves Interaction Recall@1m by

8.3 points. Decoupled+Visibility lowers Dyn Rollout error by 49.9%. Compared with Holistic-16Slot, the decoupled model improves Interaction Recall@1m from 0.643 to 0.984. The percentage deltas use the unrounded means behind Table 3. Relation tokens help when their capacity is separated from dynamic-agent capacity.

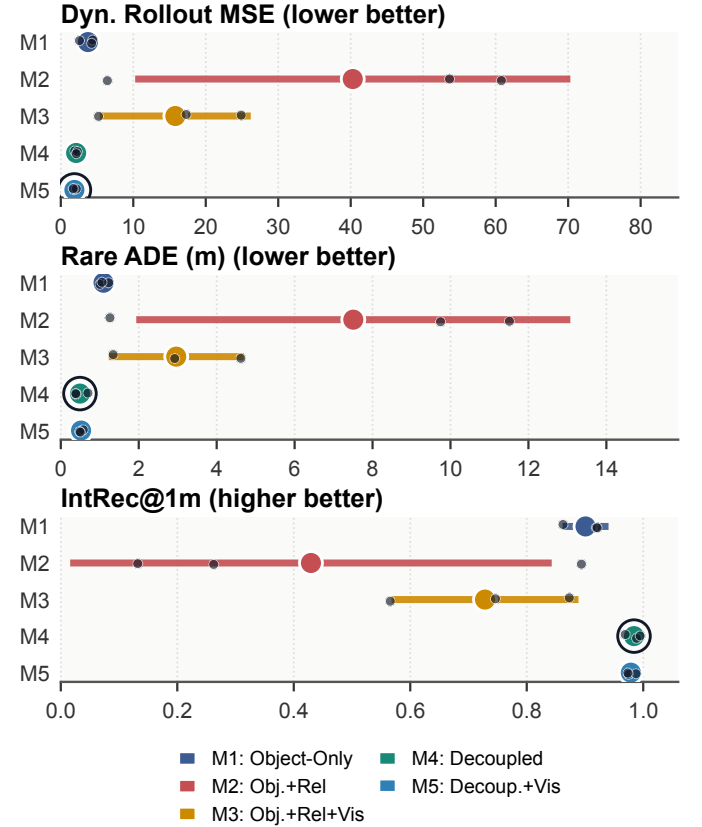


Fig. 3. Stage-0 nuScenes metrics under the fair 16-slot budget. Typed-budget variants recover dynamic rollout quality, rare-agent fidelity, and interaction recall relative to shared object-relation selection.

Naive relation mixing exposes the failure mode. Compared with Object-Only-16, Object+Relation-16 increases Dyn Rollout error from 3.74 to 40.28, increases Rare ADE from 1.10 to 7.51, and drops Interaction Recall@1m from 0.901 to 0.430. It is also unstable across seeds, with an Interaction Recall standard deviation of 0.407. The issue is shared-budget competition, not relation structure itself. Fig. 4 shows that shared-budget variants allocate too many

TABLE 4
Stage-0 typed-budget sensitivity on nuScenes. The relation-heavy 10/6 split remains usable but is weaker than 12/4.

Budget	Dyn Rollout MSE ↓	Action MSE ↓	Collision F1 ↑	Rare ADE ↓	IntRec@1m ↑
10/6	8.761 ± 1.067	0.286 ± 0.009	0.894 ± 0.007	1.574 ± 0.485	0.884 ± 0.043
12/4 main	1.876 ± 0.227	0.284 ± 0.023	0.926 ± 0.029	0.520 ± 0.049	0.979 ± 0.008

TABLE 5
Stage-1 imagination-based policy learning across datasets and nuPlan scales. Results are averaged over three seeds where repeated runs are available; $\pm$ is sample standard deviation where shown. Bold numeric entries mark the best return and collision values within each dataset block; bold condition names mark the primary trade-off condition discussed in the text. Ego stability flags divergence and is not the ranking criterion.

Dataset	Condition	Return ↑	Coll. Rate ↓	Coll. Mean ↓	Ego Stability (cos)
nuScenes	WM-Object	31.79 ± 19.70	0.597 ± 0.048	0.610 ± 0.048	0.258 ± 0.058
nuScenes	WM-Decoupled	4.34 ± 13.66	0.695 ± 0.283	0.676 ± 0.243	0.636 ± 0.108
nuScenes	WM-Decoupled-NoVis	0.34 ± 15.97	0.820 ± 0.260	0.814 ± 0.167	0.223 ± 0.033
nuPlan 20k	WM-Object	4.74 ± 13.95	0.373 ± 0.083	0.395 ± 0.057	0.426 ± 0.099
nuPlan 20k	WM-Naive-Shared	-2.66 ± 14.63	0.252 ± 0.050	0.297 ± 0.069	0.186
nuPlan 20k	WM-Decoupled	13.48 ± 4.09	0.488 ± 0.217	0.509 ± 0.193	0.546 ± 0.046
nuPlan 20k	WM-Decoupled-NoVis	17.50 ± 1.37	0.226 ± 0.105	0.251 ± 0.101	0.136 ± 0.022
nuPlan 50k	WM-Object	1.72 ± 17.89	0.610 ± 0.146	0.614 ± 0.113	0.367
nuPlan 50k	WM-Decoupled(+Vis)	-0.33 ± 4.94	0.007 ± 0.012	0.277 ± 0.111	0.255
nuPlan 50k	WM-Decoupled-NoVis	14.51 ± 2.93	0.259 ± 0.045	0.277 ± 0.033	0.222
nuPlan 50k	BC / Planner-Imit.	0.43 ± 4.23	0.327 ± 0.566	0.479 ± 0.368	0.021

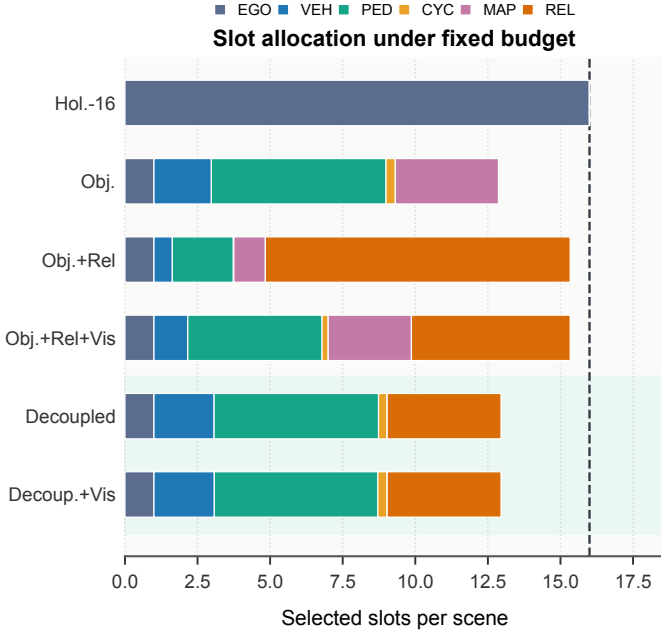


Fig. 4. Stage-0 slot-type composition. Shared object-relation selection over-allocates slots to relation tokens; typed-budget variants preserve dynamic-agent coverage.

slots to relation tokens, while decoupled variants preserve the intended dynamic-versus-relation split.

The decoupled variants lose a small amount of Collision F1 compared with Object-Only-16 (0.929 versus 0.946), likely because the fair protocol gives only four slots to relations. Typed budgeting mainly improves dynamic-agent forecasting and interaction recall; collision prediction remains competitive but is not best.

Collision F1 should not be read alone. The naive shared Object+Relation model has high Collision F1, but poor dy-

namic rollout, Rare ADE, and interaction recall. It can retain relation-level collision proxies while losing the dynamic-agent slots needed to forecast the interacting entities. For policy learning, a low-TTC relation token only helps if the relevant agents remain in the latent context. The Stage-0 claim is not that typed budgeting maximizes every auxiliary metric, but that it restores joint object-relation structure.

A non-fair 97-token holistic upper bound achieves Dyn Rollout 0.11 ± 0.12 , Collision F1 0.988 ± 0.006 , Rare ADE 0.26 ± 0.02 , and IntRec@1m 1.000 ± 0.000 . We keep it outside the main table because it uses more than six times the context of the compact variants. It only shows the remaining headroom above the fair 16-slot setting.

6.2 Stage-1 Cross-Dataset Imagination Policy Learning

Stage 0 shows that typed budgeting preserves decision-critical dynamic information under a fixed latent budget. Table 5 tests whether that representation helps policy optimization on nuScenes short-horizon imagination and nuPlan planning-oriented imagination.

Table 5 separates two effects. Better representation does not automatically imply better policy learning: on nuScenes, object-only remains the strongest main Stage-1 condition. The nuPlan ranking is different. At 20k, WM-Decoupled-NoVis achieves 17.50 ± 1.37 return versus 4.74 ± 13.95 for WM-Object and reduces collision rate from 0.373 ± 0.083 to 0.226 ± 0.105 . At 50k, it achieves 14.51 ± 2.93 return and 0.259 ± 0.045 collision rate versus 1.72 ± 17.89 and 0.610 ± 0.146 for WM-Object. The WM-Naive-Shared row connects the Stage-0 failure to Stage 1; the BC planner-imitation row gives a non-family reference.

Visibility weighting is dataset-dependent. In Stage 0, it slightly improves the best dynamic-rollout metric. In Stage 1, its role changes. On nuScenes, no-visibility is weakest among the three main variants. On nuPlan 20k and 50k,

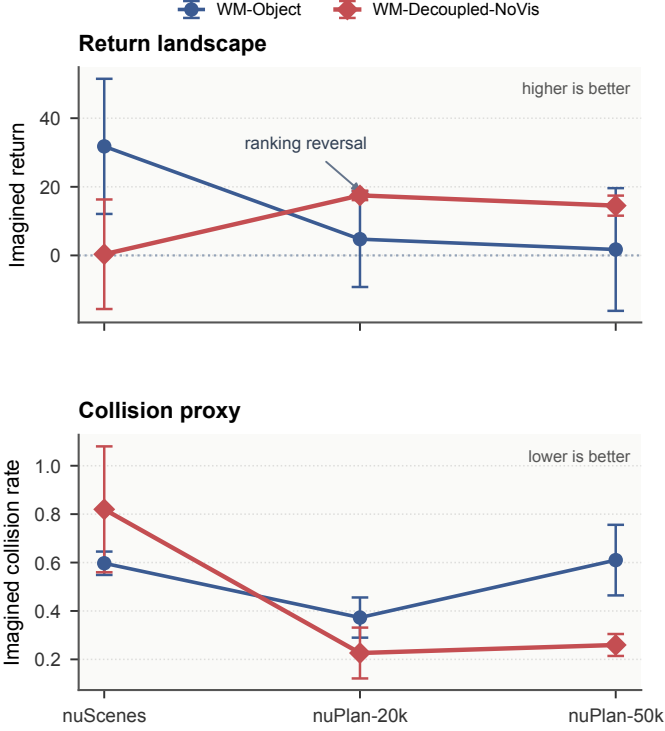


Fig. 5. Stage-1 cross-dataset evaluation. WM-Object leads on nuScenes, while WM-Decoupled-NoVis has stronger return and lower collision rate than object-only on nuPlan 20k and 50k.

no-visibility has the best return, while the visibility arm behaves more like a collision-suppression extreme. Visibility is not a universal enhancement. Reducing the relation budget from (12, 4) to (14, 2) does not rescue nuScenes Stage-1 performance, and routing relation information only to the critic gives only partial improvement. The nuScenes gap is therefore not explained by too many relation slots or by a simple actor-versus-critic fusion issue.

The nuPlan warm starts also favor decoupled abstraction. At 20k, object-only yields Dyn Rollout 168.733 and Rare ADE 17.981, whereas both decoupled variants reduce these to 3.210 and 0.848. At 50k, object-only yields Dyn Rollout 116.627, Rare ADE 17.318, and IntRec@1m 0.006, whereas the decoupled model achieves 2.438, 0.812, and 0.914. The Stage-1 nuPlan gains therefore start from a stronger representation, not from a weaker initialization. Figure 5 shows the Stage-1 ranking reversal that object-only leads on nuScenes, while WM-Decoupled-NoVis gives the strongest return and fewer collisions than object-only on nuPlan.

6.3 nuPlan Planner-Like Evaluation

Table 7 evaluates the two 50k nuPlan finalists with an offline planner-like test. This is not an external closed-loop benchmark. It reuses the nuPlan validation split and measures distance to teacher-derived planner actions while preserving the latent-rollout metrics from Stage 1.

The planner-like evaluation agrees with the 20k/50k Stage-1 tables. WM-Decoupled-NoVis has better latent return and collision metrics and is closer to the teacher-derived action target, reducing action MSE from 8.863 to 6.628. The BC / planner-imitation anchor is weaker, with

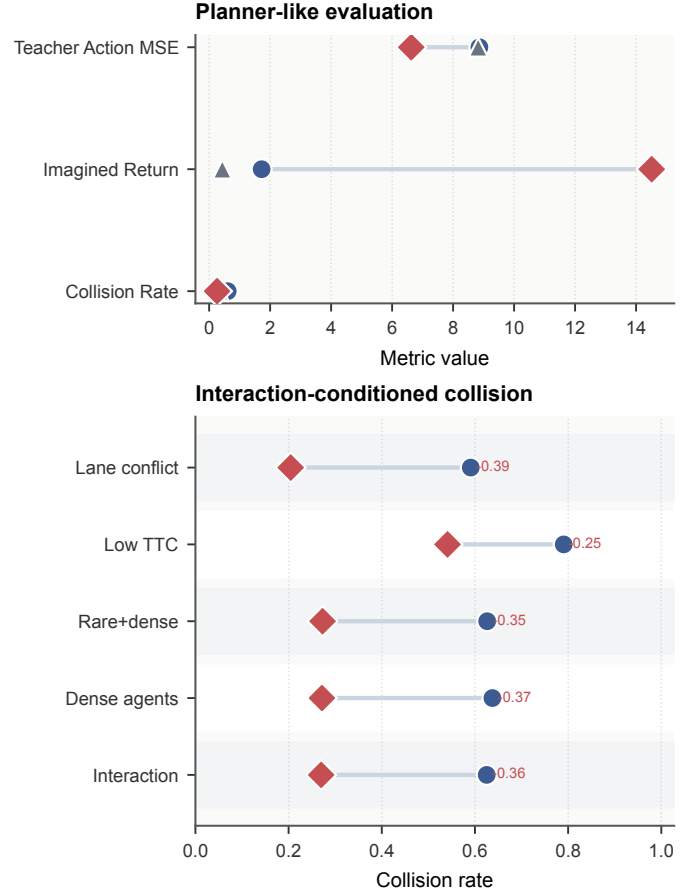


Fig. 6. nuPlan 50k offline evidence. WM-Decoupled-NoVis is closer to the teacher-derived action target and gains more on interaction-conditioned subsets.

8.824 action MSE and much lower return. WM-Object and BC show larger progress proxies, but worse action agreement and collision behavior. This is evidence for the nuPlan ranking, not a benchmark claim.

Table 8 shows where relation-aware abstraction helps. WM-Decoupled-NoVis gains most on subsets where relation structure should matter. In *lane_conflict*, action MSE drops from 7.023 to 4.225, return rises from 0.943 to 13.330, and collision rate falls from 0.591 to 0.205. The same direction holds for *low_ttc_proxy*, *rare_agent_dense*, *dense_agents*, and their union. Collision rate drops from 0.791 to 0.541 on low-TTC scenes and from 0.625 to 0.270 on the interaction-heavy union. This subset analysis is more informative than the small closed-loop smoke test because it isolates lane conflict, low TTC, and dense interactive context.

Figure 6 visualizes the same downstream pattern without adding a new benchmark. The left side emphasizes that the nuPlan winner is also closer to teacher-derived actions, while the interaction-slice view shows that its collision-rate gain is concentrated where relation information should be most useful. Figure 7 gives a complementary scene-level view: typed budgeting does not simply add relation slots; it preserves near-field dynamic coverage while reserving a small number of high-risk relation slots. Together, the two figures connect the aggregate policy result to the underlying slot-allocation mechanism.

Cross-dataset transfer and horizon sweeps give the same

TABLE 6

nuScenes Stage-1 ablations under the same 700-scene protocol. Neither a smaller relation budget nor relation-to-critic-only fusion rescues the default decoupled policy.

Condition	Return $\uparrow$	Coll. Rate $\downarrow$	Coll. Mean $\downarrow$	Ego Stability (cos)
WM-Object	31.79 ± 19.70	0.597 ± 0.048	0.610 ± 0.048	0.258 ± 0.058
WM-Decoupled	4.34 ± 13.66	0.695 ± 0.283	0.676 ± 0.243	0.636 ± 0.108
WM-Decoupled-NoVis	0.34 ± 15.97	0.820 ± 0.260	0.814 ± 0.167	0.223 ± 0.033
WM-Decoupled-14+2	2.47 ± 4.14	0.808 ± 0.196	0.774 ± 0.189	0.419 ± 0.264
WM-RelToCriticOnly	5.72 ± 7.25	0.729 ± 0.169	0.736 ± 0.163	0.453 ± 0.228

TABLE 7

nuPlan 50k offline planner-like evaluation. Results are averaged over three seeds where repeated runs are available; $\pm$ is sample standard deviation where shown. Lower action error indicates closer agreement with the teacher-derived planner action.

Condition	Teacher Action MSE $\downarrow$	Action $\Delta L_2 \downarrow$	Return $\uparrow$	Coll. Rate $\downarrow$	Progress Proxy $\uparrow$
WM-Object	8.863 ± 0.370	3.553 ± 0.118	1.722 ± 17.888	0.610 ± 0.146	2.995
WM-Decoupled-NoVis	6.628 ± 0.110	2.115 ± 0.083	14.512 ± 2.925	0.259 ± 0.045	1.082
BC / Planner-Imit.	8.824 ± 0.113	3.534 ± 0.017	0.433 ± 4.229	0.327 ± 0.566	3.000

TABLE 8

nuPlan 50k interaction-conditioned subsets. Each subset isolates scenes where relation reasoning should matter. Values are averaged over three seeds; $\pm$ is sample standard deviation where shown.

Subset	Condition	Teacher Action MSE $\downarrow$	Return $\uparrow$	Coll. Rate $\downarrow$
lane_conflict	WM-Object	7.023 ± 0.393	0.943 ± 17.975	0.591 ± 0.176
lane_conflict	WM-Decoupled-NoVis	4.225 ± 0.125	13.330 ± 3.134	0.205 ± 0.043
low_ttc_proxy	WM-Object	12.796 ± 0.425	2.650 ± 18.243	0.791 ± 0.084
low_ttc_proxy	WM-Decoupled-NoVis	11.534 ± 0.198	15.946 ± 2.851	0.541 ± 0.090
dense_agents	WM-Object	8.692 ± 0.414	1.423 ± 17.975	0.637 ± 0.170
dense_agents	WM-Decoupled-NoVis	6.340 ± 0.170	14.311 ± 3.127	0.272 ± 0.049
rare_agent_dense	WM-Object	8.729 ± 0.369	1.562 ± 17.985	0.626 ± 0.149
rare_agent_dense	WM-Decoupled-NoVis	6.494 ± 0.137	14.496 ± 3.085	0.273 ± 0.045
high_interaction_union	WM-Object	8.836 ± 0.410	1.588 ± 17.940	0.625 ± 0.150
high_interaction_union	WM-Decoupled-NoVis	6.555 ± 0.154	14.418 ± 2.994	0.270 ± 0.047

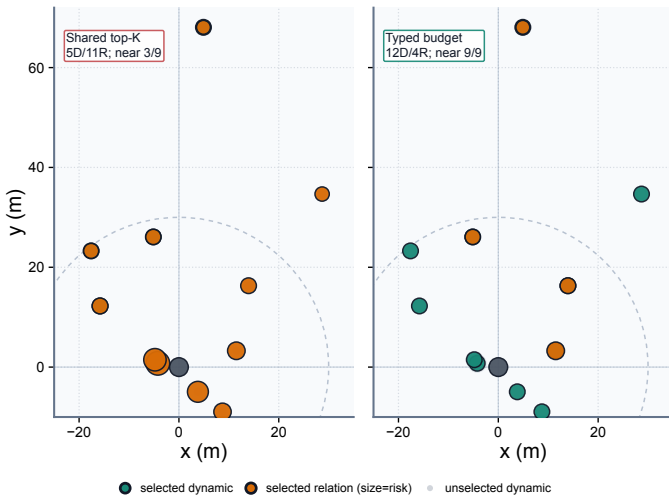


Fig. 7. Qualitative slot-selection case under the fair 16-slot budget. Bird's-eye coordinates show ego-centric token positions; all markers use the same circular shape, color denotes selected dynamic versus selected relation tokens, and relation-marker size encodes normalized interaction risk. Typed-budget selection keeps more near-field dynamic agents while reserving a small relation budget.

TABLE 9

Dataset statistics for the Stage-1 ranking reversal. Values are computed from the tokenized validation protocols.

Statistic	nuScenes	nuPlan 50k
Dynamic tokens / sample	9.715	12.164
Rare tokens / sample	2.439	3.934
Map tokens / sample	9.440	31.768
Relation tokens / sample	11.318	11.164
Dynamic visibility mean	0.746	1.000
Teacher action L_2 mean	0.539	3.420
Relation lane-conflict mean	0.111	0.116

picture. In pure cross-dataset rollout evaluation, a nuScenes-trained WM-Decoupled(+Vis) policy transfers poorly to nuPlan, dropping to -1.43 mean return with ego-step drift 0.853 . In the opposite direction, a nuPlan-trained WM-Decoupled-NoVis policy transfers to nuScenes better than a nuPlan-trained WM-Object policy, improving return from 5.51 to 16.07 and reducing collision rate from 0.624 to 0.434 . In a nuScenes horizon sweep, the decoupled collision gap grows from $+0.113$ at rollout horizon $H = 3$ to $+0.306$ at

TABLE 10

nuPlan 50k relation feature-group ablation on trained WM-Decoupled-NoVis checkpoints. The ablation is applied at evaluation time, without retraining; small differences between the full model and no-lane-priority rows are within noise.

Condition	Ablation	Teacher MSE ↓	Action ΔL_2 ↓	Return ↑	Coll. Rate ↓	Coll. Mean ↓	Stability
WM-Decoupled-NoVis	none	6.628 ± 0.110	2.115 ± 0.083	14.510 ± 2.921	0.260 ± 0.045	0.277 ± 0.033	0.222
WM-Decoupled-NoVis	no_ttc_risk	7.992 ± 0.805	3.154 ± 0.335	8.498 ± 1.397	0.895 ± 0.126	0.891 ± 0.126	0.222
WM-Decoupled-NoVis	no_lane_priority	6.627 ± 0.111	2.113 ± 0.081	14.526 ± 2.931	0.257 ± 0.052	0.274 ± 0.040	0.222
WM-Decoupled-NoVis	no_relation_semantics	7.989 ± 0.827	3.151 ± 0.351	8.724 ± 1.629	0.894 ± 0.126	0.888 ± 0.127	0.222

TABLE 11

Cross-dataset transfer and horizon sensitivity. Transfer rows report mean ± sample standard deviation over three seeds. Horizon rows come from the synchronized nuScenes sweep.

Setting	Condition	Return ↑	Coll. Rate ↓	Ego-cos step ↓	Max action norm
nuScenes → nuPlan	WM-Object	27.18 ± 11.44	0.593 ± 0.261	0.369 ± 0.203	–
nuScenes → nuPlan	WM-Decoupled(+Vis)	−1.43 ± 17.28	0.593 ± 0.320	0.853 ± 0.102	–
nuPlan → nuScenes	WM-Object	5.51 ± 11.65	0.624 ± 0.135	0.301 ± 0.154	–
nuPlan → nuScenes	WM-Decoupled-NoVis	16.07 ± 2.20	0.434 ± 0.045	0.181 ± 0.039	–
nuScenes $H = 3$	WM-Object	11.856	0.570	0.296	3.781
nuScenes $H = 3$	WM-Decoupled	1.646	0.683	0.710	4.056
nuScenes $H = 5$	WM-Object	21.814	0.603	0.190	3.846
nuScenes $H = 5$	WM-Decoupled	−1.462	0.838	0.725	4.205
nuScenes $H = 7$	WM-Object	31.950	0.610	0.143	3.850
nuScenes $H = 7$	WM-Decoupled	−5.914	0.916	0.733	4.209

$H = 7$. These checks point to compounding imagination drift on nuScenes, while the nuPlan gain survives transfer and interaction-focused slices. An oracle-token planner wrapper was also connected to the official nuPlan simulation path as an integration smoke test, but that small closed-loop run is not used as main evidence because wrapper feasibility dominates the outcome.

The dataset statistics in Table 9, relation-feature ablation in Table 10, and transfer diagnostics in Table 11 provide the mechanism-level support for this interpretation.

7 DISCUSSION

The experiments separate representation quality from policy utility, revealing that relation tokens are not universally beneficial. While Stage 0 shows typed budgeting improves decision-sufficient structure under a fixed latent budget, Stage 1 demonstrates this does not guarantee policy gains. The cross-dataset ranking reversal highlights this divergence. On nuScenes, short-horizon imagination and smaller ego-motion scales favor a simpler dynamic-only policy, making relation information harder to exploit despite latent improvements. Conversely, nuPlan provides denser dynamic context and larger teacher-action scales, allowing the decoupled abstraction to translate representation gains into policy improvements. Furthermore, the nuPlan visibility reversal shows that dataset-specific confidence signals can become distracting scale factors when transferred across protocols.

These findings challenge the common assumption that adding pairwise interaction features monotonically enriches the scene state. Under a fixed latent bottleneck, a relation feature can be harmful if it crowds out the agent state needed for action-conditioned prediction. Typed budgeting addresses this by enforcing an explicit capacity policy rather than relying on a shared top- K pool or hand-crafted distance rules. By requiring a minimum dynamic-agent

inventory before adding relation evidence, the method preserves complementary token families. This makes the object-relational abstraction a highly effective intermediate representation between perception and planning, where strict latency and memory budgets demand a compact yet descriptive scene state. The two-stage protocol is crucial here, exposing whether the representation is sufficient and whether the policy can actually exploit that sufficiency.

The study has several limitations that motivate future work. First, relying on oracle tokens excludes perception noise, suggesting a need for perception-aware evaluation where token uncertainty is explicitly modeled. Second, the policy claims rest on imagination-based training rather than large-scale closed-loop deployment. Future iterations must connect this typed abstraction to official simulator protocols with matched controller logic. Third, the role of visibility remains unresolved, acting variably as an enhancement, a stabilizer, or a liability depending on the dataset. Finally, replacing the fixed (12, 4) allocation with adaptive budgeting could allow the model to predict scene-level capacity needs while respecting global latency constraints.

8 CONCLUSION

This paper introduces typed budgeting to manage latent capacity bottlenecks in autonomous driving world models. Partitioning the context into dynamic-agent and relation capacities prevents high-scoring interaction features from crowding out essential agent states. Our evaluation reveals that improving latent representation quality does not guarantee downstream policy gains. Instead, relation token utility depends heavily on dataset density, action scale, and task horizon. This challenges the assumption that adding pairwise interactions is universally beneficial under fixed memory constraints. The proposed abstraction provides a robust intermediate representation, balancing descriptive power with the strict latency and memory requirements of modern autonomy stacks.

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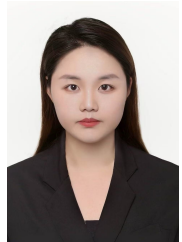


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